

STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	1-3 Semester (2025-26)
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Branch	CSE
Course Outcome	1

Case Study Topic: Hacker News Story Ranking using AVL Tree

Word Done Summary:

In this case study, we analysed how Hacker News maintains story rankings using an AVL Tree. The stories are ranked based on scores in descending order. We studied why directly modifying a node score inside an AVL Tree is unsafe because it violates the BST property. We also examined the correct approach of deleting and reinserting nodes whenever scores are updated. Finally, we calculated the operational workload and analysed whether the AVL operations can handle the given upvote traffic efficiently.

Implementation Details:

1. Studied AVL Tree structure used for ranking stories.
2. Analysed BST invariant violations caused by in-place key mutation.
3. Implemented delete-and-reinsert strategy for score updates.
4. Calculated time complexity of AVL operations.
5. Computed operations per second for Hacker News workload.
6. Evaluated whether a single CPU core can handle the workload efficiently.

Result:

The AVL Tree efficiently maintains ranked story ordering while supporting dynamic score updates. The delete-and-reinsert strategy preserves AVL balancing and BST invariants without causing incorrect node positioning.

Output:

```
Hacker News Story Rankings:
Tech Update : 30
AI News : 50
Programming Tips : 60
Startup Story : 70
Open Source : 90

Score Update:
Deleting old score and reinserting updated score.

Updated Rankings:
Tech Update : 30
AI News : 50
Programming Tips : 60
Startup Story : 70
Open Source : 90
Open Source : 95
```

GitHub Link:

<https://github.com/ketanasindhusomisetty/DSA-Case-Studies/tree/main>

Student Signature	Faculty Signature
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